Python2

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Logical operators

1) And → x and y → x&y
2) or → x or y → x|y
3) not → not x → ~x
4) xor → x^y

And

x	y	
1	1	1
1	0	0
0	1	0
0	0	0

or

x	y	
1	1	1
1	0	1
0	1	1
0	0	0

not

x	
1	0
0	1

XOR

x	y	
1	1	0
1	0	1
0	1	1
0	0	0

Bitwise operator

&
|
^
~
<<
>>

x = 2
y = 7

8bits

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	0	0	0	0	0	1	0

→ 0010

64 32 16 8 4 2 1

0000111 → 0111

12 0001100 → 1100

x = 2 = 0010
y = 7 = 0111 &
0010 → 2

$y = 7 = 0111$

$\& \quad 0010 \rightarrow 2$

$x \mid y \quad x = 0010$

$y = 0111$

$0111 \rightarrow 7$

$x \wedge y = x = 0010$

$y = 0111$

$0101 \rightarrow 5$

$\sim x \Rightarrow$ change the sign adding of one

$x = 2 \Rightarrow \sim x = -(2+1) \Rightarrow -3$

$\sim y \Rightarrow -(7+1)$

$\Rightarrow -8 \rightarrow R$

$x << 2 = 0010$

32 16 8 4 2 1

$001000 \Rightarrow 8$

000010

$001000 \rightarrow$

$010000 \rightarrow 16$

$x << 3 \Rightarrow$

$x = 000010$

32 16 8 4 2 1

$010000 \rightarrow 16$

$\rightarrow R$

$x >> 1 = 0010$

$0001 \rightarrow 1$

$x >> 2 = 0010$

$0000 \rightarrow 0$

Identity operator $\Rightarrow$ is

$x_1 = 5 \quad y_1 = 5 \quad x_1$ is $y_1 \Rightarrow$ True

$x_2 =$ 'Hello' $\quad y_2 =$ 'Hello' $\quad x_2$ is $y_2 \Rightarrow$ True

$x_3 =$ 'Hello', $\quad y_3 =$ 'h' $\quad x_3$ is $y_3 \Rightarrow$ False

$x_4 = [1, 2, 3] \quad y_4 = [1, 2, 3] \quad x_4$ is $y_4 \Rightarrow$ False

list one

list one

$x_4 = [1, 2, 3]$ ... list one

list one

Assignment operators

$x = x + 1$ ✓ $y = 2$ $x \boxed{2}_3$

$x \mathrel{+}= 1$

$x \mathrel{-}= 2 \Rightarrow x = x - 2$ $x \boxed{3}_1$ 2

$x \mathrel{*}= 2 \Rightarrow x = x * 2$

$x \mathrel{|}= y \Rightarrow x = x | y$

$x \mathrel{||}= y \Rightarrow x = x || y$

$x \mathrel{**}= y$ $x = x ** y$

16 8

0 0 0 0

0 1 0 0

or 0 1 0 0 ⇒ 4

$x << 2$ 1 0 0 0 0 ⇒ 16

membership operator ⇒ 'in'

$x = [2, 3, 4, 7, 12]$

4 in x ⇒ True

6 in x ⇒ False